MIDLANDS STATE UNIVERSITY

FACULTY OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF APPLIED MATHEMATICS & STATISTICS

HCSE233/HMAT133: LINEAR ALGEBRA

SESSIONAL EXAMINATIONS

JUNE 2024

DURATION: 3 HOURS

INSTRUCTIONS

1. *Answer **(ALL)** Questions From Section A and any **THREE (3)** from Section B*
2. *All questions carry marks as indicated.*

SECTION A: 40 MARKS.

A1 A real matrix is orthogonal iff $A^T A = AA^T = I$. Show that the matrix $A = \begin{pmatrix} sin\theta & cos\theta \\ cos\theta & -sin\theta \end{pmatrix}$ is orthogonal [7]

A2 a) Express $(0, 1, 1)$ as a linear combination of the following vectors $(2, 1, -1), (0, 0, 1)$ and $(2, 0, 0)$.

b) Let $M = \begin{pmatrix} 1 & 1 & 1 \\ 2 & -3 & -1 \\ 1 & 2 & 2 \end{pmatrix}$.

i) Find the determinant and inverse of M.

ii) Use the method of inverses to solve the system $MX = (4\ 1\ 5)^T$, where $X = (x\ y\ z)^T$.

[4,5,6]

A3 Find the symmetric matrix B which corresponds to the quadratic forms:

a) $q(x, y) = 4x^2 + 5xy - 7y^2$,

b) $q(x, y, z) = 3x^2 + 4xy - y^2 + 8xz - 6yz + z^2$.

[3,3]

A4 Find an orthonormal basis for the subspace U of $\Re^4$ spanned by the vectors

$$v_1 = (1, 1, 1, 1), v_2 = (1, 1, 2, 4), v_3 = (1, 2, -4, -3$$

[5]

A5. Let $A = \begin{pmatrix} 2 & 3 & 1 \\ 0 & 4 & -1 \\ -3 & -6 & -3 \end{pmatrix}$ Show that $B = \begin{pmatrix} 2 \\ 3 \\ -3 \end{pmatrix}$ is an eigenvector of A and find the corresponding eigenvalue.

[4,3]

SECTION B: 60 MARKS.

B6 Let A=$\begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$. Find:

a) all eigenvalues of A and the corresponding eigenspaces.

b) an invertible matrix P such that $D = P^{-1}AP$ is diagonal.

c) A^5 and $f(A)$ where $f(t) = t^4 - 3t^3 - 7t^2 + 6t - 15$.

[12,3,5]

B7 Use the Gauss elimination method to solve the following system of equations

$$\begin{aligned} x + y + z &= 4, \\ 2x - 3y - z &= 1, \\ x + 2y + 2z &= 5. \end{aligned}$$

Verify your solution using Crammer's rule.

[12,8]

B8 a) Consider the basis $B = \{v_1, v_2, v_3\}$ where $v_1 = (1, 0, 1)$, $v_2 = (1, 1, 0)$, $v_3 = (1, 0, 0)$. Let $T : R^3 \to R^2$ be a linear transformation such that $T(v_1) = (1, 0)$, $T(v_2) = (2, -1)$, $T(v_3) = (2, 3)$. Find $T(1, 3, 6)$.

b) Let P_3 be the vector space of polynomials of degree 3 and less with real coefficients with the basis $\{1, t, t^2, t^3\}$. $T : P_3 \to P_3$ is a linear transformation such that $T(1) = 1 + t^2$, $T(t) = t$, $T(t^2) = 2 + 3t^2$, $T(t^3) = 1 - 2t + t^3$. Compute $T(3 - 6t + 2t^2 - 5t^3)$.

[10,10]

B9 Consider the vector space V of polynomials f(t) of degree ≤ 2 with inner product $\langle f, g\rangle = \int_{-1}^{1} f(t)g(t)dt$.

a) Find $\langle f, g\rangle$ where $f(t) = t + 2$ and $g(t) = t^2 - 3t + 4$.

b) Find the matrix A of the inner product with respect to the basis $\{1, t, t^2\}$ of V.

c) Show that $\langle f, g\rangle = [f]^T A[g]$ with respect to the basis $\{1, t, t^2\}$ of V.

[9,3,8]

END OF QUESTION PAPER